ChannelShuffle#

https://pytorch.org/docs/stable/generated/torch.nn.ChannelShuffle.html#torch

Description:

Divides and rearranges the channels in a tensor. This operation divides the channels in a tensor of shape (N,C,∗)(N, C, *)(N,C,∗) into g groups as (N,Cg,g,∗)(N, \frac{C}{g}, g, *)(N,gC,g,∗) and shuffles them, while retaining the original tensor shape in the final output. groups (int) – number of groups to divide channels in. Return the extra representation of the module.

Function Signature

```
class torch.nn.ChannelShuffle(groups)[source]#
Parameters
extra_repr()[source]#
Return type
forward(input)[source]#
```

Returns:

Tensor

Code Examples

```
>>> channel_shuffle = nn.ChannelShuffle(2)
>>> input = torch.arange(1, 17, dtype=torch.float32).view(1, 4,
2, 2)
>>> input
tensor([[[[ 1.,  2.],
          [ 3.,  4.]],
         [[ 5.,  6.],
          [ 7.,  8.]],
         [[ 9., 10.],
          [11., 12.]],
         [[13., 14.],
          [15., 16.]]]])
>>> output = channel_shuffle(input)
>>> output
tensor([[[[ 1.,  2.],
          [ 3.,  4.]],
         [[ 9., 10.],
          [11., 12.]],
         [[ 5.,  6.],
          [ 7.,  8.]],
         [[13., 14.],
          [15., 16.]]]])
```

Detailed Documentation

Documentation

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groups (int) – number of groups to divide channels in.

Return the extra representation of the module.

Runs the forward pass.

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